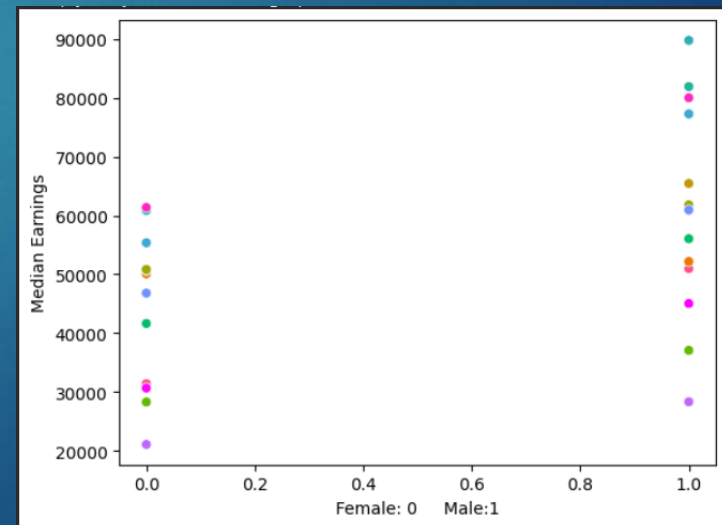
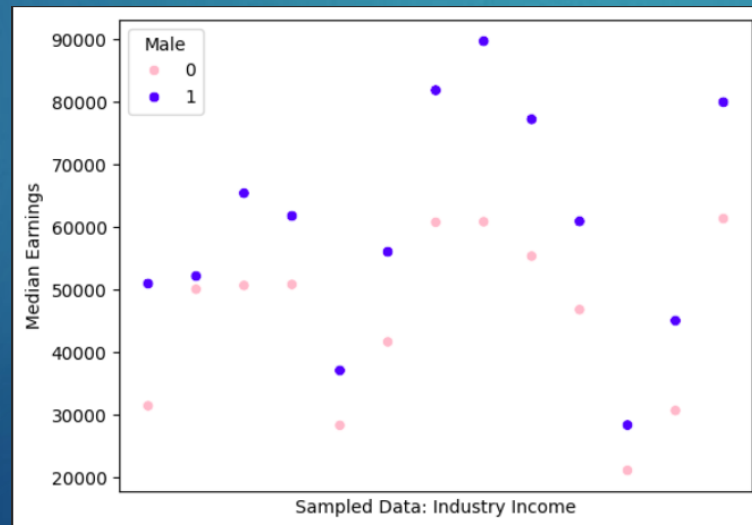


ECON Final

BRAD PETARS

Research Question

- ▶ Does the gender pay gap exist, and does controlling for industry influence it?
- ▶ Null Hypothesis 1: Being male has no effect on median earnings
- ▶ Equation 1: $\text{Median_Wages} = \beta_0 + (\beta_1 * \text{Male}) + \text{epsilon}$
- ▶ Null Hypothesis 2: Holding industry constant, being male has no effect of median earnings
- ▶ Equation 2: $\text{Median_Wages} = \beta_0 + (\beta_1 * \text{Male}) + (\beta_2 * \text{Industry}) + \text{epsilon}$



Key Findings + Conclusion

► Equation 1:

- Intercept (β_0): ~\$45k
 - A female can expect to make about \$45k a year in average median earnings
- Male Coefficient (β_1): ~\$15k
 - Being male implies an extra \$15k in median annual earnings
 - P-Value: 0.025 → does not pass 99% confidence level

► Equation 2:

- Intercept (β_0): ~\$33k
 - A female can expect to make about \$33k a year in average median earnings
- Male Coefficient (β_1): ~\$15k
 - Being male implies an extra \$15k in median annual earnings (still)
 - P-Value: 0.000 → passes 99% confidence level → reject the null

- Conclusion: With a 99% confidence level, we can reject the null hypothesis that, holding industry constant, being male does not have an influence on median earnings

